

README.md & Strategic Execution Plan

Lightweight MobileViT-Graph Network for Topological Rural Road Extraction from Satellite Imagery

1. Project Overview & Rationale

In developing nations like India, the infrastructure audit and mapping required under initiatives such as the Pradhan Mantri Gram Sadak Yojana (PMGSY) drive an urgent need to accurately map millions of kilometers of rural, unpaved roads. Current manual auditing workflows are prohibitively labor-intensive and slow. While standard deep learning semantic segmentation models have excelled at urban highway mapping, they exhibit severe performance drop-offs (Domain Shift) when deployed in rural environments.

Rural unpaved roads are typically slender, highly irregular, and frequently obstructed by dense tree canopies, building shadows, and complex topographical terrain. Standard segmentation models rely heavily on volumetric pixel overlap metrics like Binary Cross-Entropy (BCE) or standard Intersection-over-Union (IoU), which optimize for spatial pixel counts rather than functional connectivity. This causes fragmented outputs and broken road topologies.

The Core Project Objective: To construct an ultra-lightweight encoder-decoder architecture utilizing **MobileViT v2** as a fast, low-parameter backbone, synthesized with graph-theoretic/topology-preserving constraints (**Centerline-Dice Loss** / **clDice**) to enforce contiguous, navigable road predictions on Indian rural topographies while optimizing for edge-device deployment.

2. Technical Approach & Pipeline

The pipeline overcomes the standard $O(N^2)$ computational complexity of heavy Vision Transformers (ViTs) while capturing long-range global context to "see" past occlusions where traditional CNNs fail.

- **Backbone Encoder:** MobileViT v2 replaces local processing in convolutions with global processing using localized Transformers, dramatically lowering the parameter footprint while maintaining a massive receptive field.
- **Loss Function:** The network replaces standard pixel-wise spatial losses with graph-theoretic **clDice Loss**. clDice extracts the morphological skeleton of both the ground-truth centerline and the predicted road mask, calculating overlap explicitly on the topological graph:

$$clDice(X, Y) = 2 \times \frac{1}{2}(Prec_{cl}(X, Y) + Rec_{cl}(X, Y))$$

- **Weak Supervision Strategy:** To eliminate manual annotating bottlenecks, training utilizes 1-pixel-wide vector line-strings extracted from OpenStreetMap (OSM) centerlines as weak labels, utilizing clDice to iteratively expand and map precise road boundaries.

3. 4-Member Project Structure & Core Roles

To execute this project seamlessly using the foundational **DeepGlobe Dataset** and targeted Indian Rural Satellite imagery, the engineering responsibilities are structured across 4 distinct, highly collaborative roles:

Role	Primary Domain & Responsibilities	Key Interfaces & Handoffs
Member 1 LEAD ARCHITECT	Designing and building the core lightweight encoder-decoder architecture using MobileViT v2 . Implementing localized self-attention blocks and multi-scale feature aggregators. Optimizing parameter count, compute profiles, and inference speed (FPS).	Outputs the model backbone to Member 2 for loss incorporation, and delivers the compiled model to Member 4 for evaluation.
Member 2 GRAPH & LOSS ENGINEER	Implementing the graph-theoretic clDice (Centerline-Dice) loss function , morphological skeletonization layers, and integrating them into the model pipeline. Constructing topology-preserving loss terms and structural continuity graphs.	Integrates loss routines directly into Member 1's architecture. Delivers graph-constrained predictions to Member 4 for tracking metric compliance.
Member 3 DATA & PIPELINE ENGINEER	Preprocessing and handling the DeepGlobe Dataset . Building the data loading and augmentation pipeline. Implementing weak supervision layers by aligning OpenStreetMap (OSM) centerlines with satellite frames. Managing domain adaptation and occlusion masks.	Provides clean, augmented batched datasets and aligned vector/pixel masks to Member 1 and Member 2 for training cycles.
Member 4 VALIDATION & MLOPS	Building advanced validation suites specializing in non-pixel routing metrics, specifically APLS (Average Path Length Similarity) and TOPO metrics. Managing ablation studies, tracking convergence profiles, and testing real-world deployment parameters (FPS, Latency).	Consumes models and predictions from Members 1 & 2 . Provides critical iteration feedback and final comparative benchmark performance logs.

4. Phased Execution WBS (6-Week Timeline)

Phase 1: Environment Baseline & Data Pipeline (Week 1)

- **Member 1:** Initialize the repository framework, baseline U-Net structures, and verify the deep learning framework stack (PyTorch/TensorFlow).
- **Member 2:** Research and code a standalone prototype of the morphological skeletonization algorithm using differentiable operations.
- **Member 3:** Ingest the DeepGlobe dataset. Design the tiling, normalization, and geometric transformation pipelines to handle high-resolution imagery.
- **Member 4:** Stand up the tracking dashboard (e.g., Weights & Biases) and establish basic validation benchmarks using standard IoU and F1-score scripts.

Phase 2: Core Model Formulation & Loss Fusion (Weeks 2-3)

- **Member 1:** Construct the **MobileViT v2 encoder** blocks. Integrate localized transformer modules to replace standard heavy self-attention layers, keeping structural parameters ultra-low.
- **Member 2:** Implement the full **clDice loss function**. Build the graph-theoretic constraints that calculate precision and recall on predicted and ground-truth centerline skeletons.
- **Member 3:** Build the weak supervision pipeline. Set up scripts to ingest open-source OpenStreetMap (OSM) centerlines and rasterize them into 1-pixel line strings for weakly-supervised training routines.
- **Member 4:** Implement the **APLS (Average Path Length Similarity)** metric library, translating raw model masks into functional connectivity graphs for evaluation.

Phase 3: Integration, Training & Domain Tuning (Week 4)

- **Member 1 & 2:** Fuse the MobileViT v2 backbone with the clDice loss routine. Run initial training sweeps on the baseline DeepGlobe dataset.
- **Member 3:** Introduce advanced augmentation layers (shadow simulations, canopy occlusions, random color jitters) to simulate challenging Indian rural environments.
- **Member 4:** Run comprehensive model validation sweeps. Generate loss convergence logs and map network graph layouts against true paths.

Phase 4: Optimization, Ablation & Final Delivery (Weeks 5-6)

- **Member 1:** Perform lightweight structural tuning and export optimized formats (e.g., ONNX, TensorRT) to measure real-world inference frames per second (FPS).
- **Member 2:** Fine-tune the multi-loss balance coefficients matching spatial Cross-Entropy and structural clDice values.
- **Member 3:** Execute the final test suites evaluating cross-region domain adaptation (e.g., DeepGlobe urban to Indian rural imagery shifts).
- **Member 4:** Aggregate final outcomes into comprehensive ablation tables detailing parameter reduction, topological similarity, and APLS scores against heavy 100M parameter Transformer baselines. Draft the project's technical documentation and finalized codebase summary.